PART 1: Calibration (CAL) Standards

This initial phase established the baseline references for the hardware validation.

Table 1 & 2: System Calibration

Parameter	Paraphrased Description	Measured Value
Peak-to-Peak Voltage	The maximum voltage swing of the CAL output.	1.01 V
Output Period	The time duration of one full cycle.	1 ms
Output Frequency	The number of cycles per second.	1 kHz

PART 2.1: Signal Generator Characterization

This section details the performance of various waveforms across different frequencies.

Table 1: Signal Output Measurements

Waveform Type	Voltage Level	Cycle Duration	Measured Frequency
2kHz SINE	1.96 Vpp	478.4 μs	2.09 kHz
100kHz COS	4.15 Vpp	10.03 μs	99.67 kHz
100kHz SINE	3.83 Vpp	9.98 μs	100.02 kHz

Note: A comparative analysis was performed to identify discrepancies between the SINE and COS outputs.

PART 2.2: Gain and Input Scaling

Testing how the hardware responds to specific input levels and how the gain varies between maximum and minimum settings.

Table 3 & 4: Voltage Transformation

- **Initial Test:** An input of **3.92 Vpp** resulted in a **4.11 V** output.
- **Secondary Test:** Using the same input parameters, the hardware produced a significantly lower output of **360 mV**.

Gain Analysis (Tables 1 & 2)

Input Source	Setting	Input Voltage	Output Voltage	Calculated Gain
Input A	Maximum	3.79 V	4.11 V	1.08
Input A	Minimum	3.79 V	80 mV	0.02
Input B	Maximum	3.79 V*	4.07 V	1.07
Input B	Minimum	3.79 V*	80 mV	0.02
*Input B values referenced from Table 1.				

PART 3: Summing and Frequency Oscillation

Focuses on the output of the adder module and the operational ranges of the Local Oscillator.

- **Adder Output:** The combined signal voltage reached **7.71 V**.
- **Sine Component:** The individual SINE output was measured at **4.44 V**.

Table 2: Frequency Tuning Ranges

Oscillator Setting	Lowest Frequency	Highest Frequency
LO (Low) Range	1.137 kHz	18.26 kHz
HI (High) Range	55.65 kHz	139.3 kHz

PART 4: Modulation Validation

Final analysis of the Amplitude Modulation (AM) characteristics.

- **Modulation Voltage:** A **5.49 V** input was stepped up to a **13.1 V** output.
- **Low-Level Output:** A secondary measurement recorded a minimal output of **50 mV**.

Table 1: Modulation Index Parameters

P Dimension	Q Dimension	m (Modulation Index)
4.67 V	2.4 V	0.321